EXP 1: Simulation-Based Electromagnetic Analysis of TEM, TE and TM Propagation Modes in Microwave Transmission Structures

Objectives of this Experiment:

1. To develop electromagnetic simulation models for TEM, TE and TM propagation modes in suitable microwave transmission structures.

2. To visualize and analyze the electric-field and magnetic-field distributions associated with different propagation modes.

3. To determine the cutoff frequency, propagation constant and field characteristics of selected TE

and TM modes.

4. To compare the propagation behavior of different modes and identify their suitability for practical microwave transmission.

CODE:

```
clc;
clear;
close all;
%% EXPERIMENT 1: TEM, TE AND TM PROPAGATION MODES
% WR-90 Rectangular Waveguide
%% 1. INPUT PARAMETERS
c = 3e8;
f = 10e9;
a = 22.86e-3;
b = 10.16e-3;
L = 100e-3;
k0 = 2*pi*f/c;
```

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%% 2. THEORETICAL CUTOFF FREQUENCIES
fc_TE10 = c/(2*a);
fc_TM11 = (c/2)*sqrt((1/a)^2 + (1/b)^2);
%% 3. PROPAGATION CONSTANTS
beta_TE10 = k0*sqrt(1-(fc_TE10/f)^2);
if f > fc_TM11
   beta_TM11 = k0*sqrt(1-(fc_TM11/f)^2);
else
   beta_TM11 = NaN;
   alpha_TM11 = k0*sqrt((fc_TM11/f)^2-1);
end
%% 4. DISPLAY RESULTS
fprintf('\nMICROWAVE ENGINEERING - EXPERIMENT 1\n');
fprintf('-----------------------------------\n');
fprintf('Operating frequency = %.2f GHz\n',f/1e9);
fprintf('Width a = %.2f mm\n',a*1000);
fprintf('Height b = %.2f mm\n',b*1000);
fprintf('Length L = %.2f mm\n\n',L*1000);
fprintf('TE10 cutoff frequency = %.4f GHz\n',fc_TE10/1e9);
fprintf('TM11 cutoff frequency = %.4f GHz\n',fc_TM11/1e9);
fprintf('TE10 propagation constant = %.3f rad/m\n',beta_TE10);
if isnan(beta_TM11)
   fprintf('TM11 is below cutoff at 10 GHz.\n');
   fprintf('TM11 evanescent decay constant = %.3f Np/m\n', ...
      alpha_TM11);
else
   fprintf('TM11 propagation constant = %.3f rad/m\n',beta_TM11);
```

```
end
%% 5. FIELD DISTRIBUTIONS
x = linspace(0,a,100);
y = linspace(0,b,80);
[X,Y] = meshgrid(x,y);
% TE10 mode
E_TE10 = sin(pi*X/a);
H_TE10 = cos(pi*X/a);
% TM11 mode
Ez_TM11 = sin(pi*X/a).*sin(pi*Y/b);
Hx_TM11 = sin(pi*X/a).*cos(pi*Y/b);
Hy_TM11 = cos(pi*X/a).*sin(pi*Y/b);
H_TM11 = sqrt(Hx_TM11.^2 + Hy_TM11.^2);
H_TM11 = H_TM11/max(H_TM11(:));
%% GRAPH 1 TO GRAPH 7 USING SUBPLOT
figure('Color','w','Position',[50 50 1400 850]);
%% GRAPH 1: TE10 ELECTRIC FIELD
subplot(3,3,1);
surf(X*1000,Y*1000,E_TE10);
shading interp;
colorbar;
xlabel('Width x (mm)');
ylabel('Height y (mm)');
zlabel('Normalized E_y');
title('TE_{10} Electric Field');
view(45,30);
grid on;
```

```
%% GRAPH 2: TE10 MAGNETIC FIELD
subplot(3,3,2);
surf(X*1000,Y*1000,H_TE10);
shading interp;
colorbar;
xlabel('Width x (mm)');
ylabel('Height y (mm)');
zlabel('Normalized H_z');
title('TE_{10} Magnetic Field');
view(45,30);
grid on;
%% GRAPH 3: TM11 ELECTRIC FIELD
subplot(3,3,3);
surf(X*1000,Y*1000,Ez_TM11);
shading interp;
colorbar;
xlabel('Width x (mm)');
ylabel('Height y (mm)');
zlabel('Normalized E_z');
title('TM_{11} Electric Field');
view(45,30);
grid on;
%% GRAPH 4: TM11 MAGNETIC FIELD
subplot(3,3,4);
surf(X*1000,Y*1000,H_TM11);
shading interp;
colorbar;
```

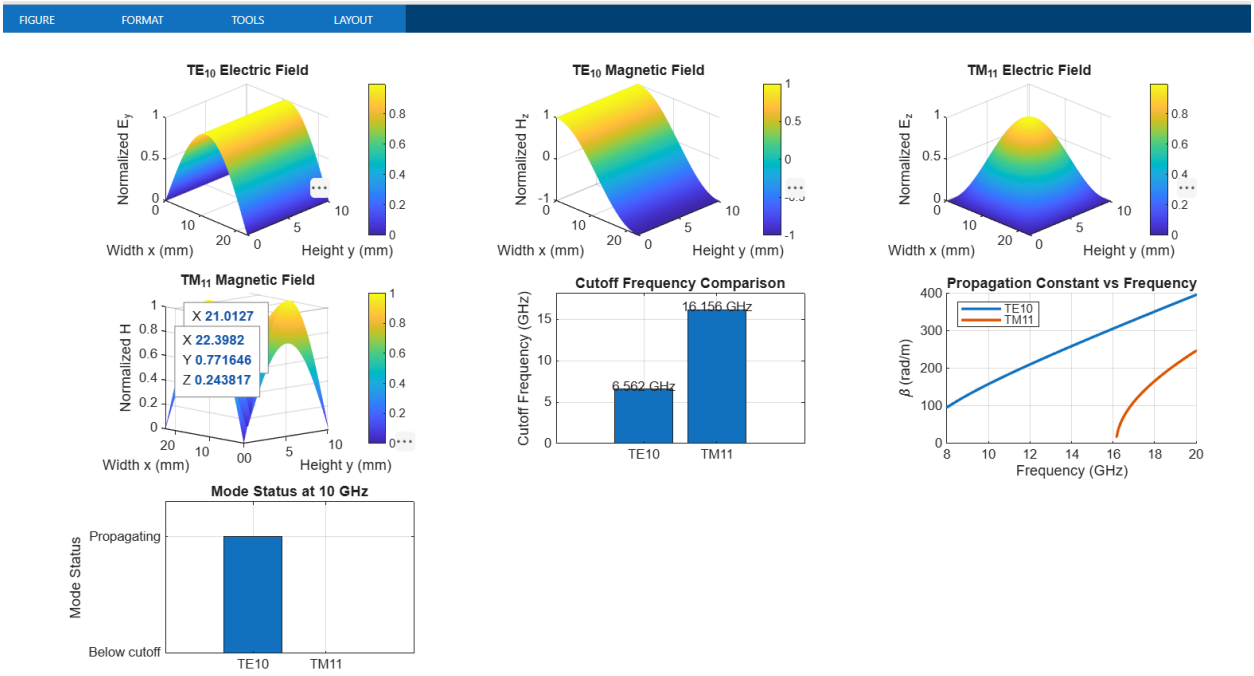
```
xlabel('Width x (mm)');
ylabel('Height y (mm)');
zlabel('Normalized H');
title('TM_{11} Magnetic Field');
view(45,30);
grid on;
%% GRAPH 5: CUTOFF FREQUENCY COMPARISON
subplot(3,3,5);
cutoffGHz = [fc_TE10 fc_TM11]/1e9;
bar(cutoffGHz);
set(gca,'XTick',1:2);
set(gca,'XTickLabel',{'TE10','TM11'});
ylabel('Cutoff Frequency (GHz)');
title('Cutoff Frequency Comparison');
grid on;
for i = 1:2
   text(i,cutoffGHz(i)+0.3, ...
      sprintf('%.3f GHz',cutoffGHz(i)), ...
      'HorizontalAlignment','center');
end
ylim([0 max(cutoffGHz)+2]);
%% GRAPH 6: PROPAGATION CONSTANT VS FREQUENCY
subplot(3,3,6);
freq = linspace(8e9,20e9,500);
fc_modes = [fc_TE10 fc_TM11];
labels = {'TE10','TM11'};
```

```
hold on;
for i = 1:2
    beta = nan(size(freq));
    valid = freq > fc_modes(i);
    beta(valid) = (2*pi*freq(valid)/c).* ...
        sqrt(1-(fc_modes(i)./freq(valid)).^2);

    plot(freq/1e9,beta,'LineWidth',2);
end
xlabel('Frequency (GHz)');
ylabel('\beta (rad/m)');
title('Propagation Constant vs Frequency');
legend(labels,'Location','northwest');
grid on;
xlim([8 20]);
hold off;
%% GRAPH 7: MODE PROPAGATION STATUS
subplot(3,3,7);
status = [f > fc_TE10, f > fc_TM11];
bar(double(status));
set(gca,'XTick',1:2);
set(gca,'XTickLabel',{'TE10','TM11'});
set(gca,'YTick',[0 1]);
set(gca,'YTickLabel',{'Below cutoff','Propagating'});
ylabel('Mode Status');
title('Mode Status at 10 GHz');
ylim([0 1.3]);
```

```
grid on;
%% 8. PERCENTAGE DEVIATION
fc_sim_GHz = NaN;
if ~isnan(fc_sim_GHz)
    deviation = abs(fc_sim_GHz-fc_TE10/1e9) ...
        /(fc_TE10/1e9)*100;
    fprintf('\nTE10 cutoff frequency deviation = %.3f%%\n', ...
        deviation);
else
    fprintf('\nCutoff deviation not calculated:\n');
    fprintf('Enter the actual simulated cutoff frequency.\n');
end
%% 9. CONCLUSION
fprintf('\nCONCLUSION\n');
fprintf('TE10 propagates at 10 GHz because f > fc.\n');
fprintf('TM11 is below cutoff because f < fc.\n');
fprintf('All field graphs are idealized normalized patterns.\n');
```

OUTPUT:


FIGURE
FORMAT
TOOLS
LAYOUT
TE10 Electric Field
Normalized Ey
Width x (mm)
Height y (mm)
TE10 Magnetic Field
Normalized Hz
Width x (mm)
Height y (mm)
TM11 Electric Field
Normalized Ez
Width x (mm)
Height y (mm)
TM11 Magnetic Field
Normalized H
X 21.0127
X 22.3982
Y 0.771646
Z 0.243817
Width x (mm)
Height y (mm)
Cutoff Frequency Comparison
Cutoff Frequency (GHz)
16.156 GHz
6.562 GHz
TE10
TM11
Propagation Constant vs Frequency
β (rad/m)
TE10
TM11
Frequency (GHz)
Mode Status at 10 GHz
Mode Status
Propagating
Below cutoff
TE10
TM11